

Jieping Luo

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EDUCATION

M.Sc. in Statistical Science, University of Oxford (2025-2027)

Expected First class

- **Modules include:** Statistical Inference, Advanced Statistics and Machine Learning, Applied and Computational Statistics, Bayesian Methods, Graphical Models.

B.Sc. in Statistics, Hong Kong Baptist University (2021-2025)

GPA: 3.94/4.00 | Rank: 1/94

- **Modules include:** Advanced Probability and Statistics, Applied Stochastic Processes, Bayesian Statistics, Discrete Structures, Natural Language Processing, Deep Learning, Interdisciplinary Mathematical Modelling.
- **Honours:** Outstanding Graduate Scholarship (2025); National Scholarship (2023–2024); President's Recommendation Scholarship (2022–2024); Dean's Recommendation Scholarship (2021–2022); First Class Scholarship (2021-2025).
- **Awards:** Meritorious Winner, Mathematical Contest in Modeling (MCM); Second Prize, Guangdong Division, China Undergraduate Mathematical Contest in Modeling; Global Top 20, FinTech Olympiad.

RESEARCH EXPERIENCE

Optimal Sequential E-values for Correlated Bernoulli Data

2026 – Present

Master's Dissertation

- Extended optimal sequential e-value theory from independent to correlated multivariate Bernoulli observations.
- Derived growth-optimal e-values and reverse information projections under composite null hypotheses with sum constraints.
- Studied the effect of correlation on asymptotic growth rate, expected stopping time and finite-sample power.
- Developed exact bivariate results and general- d local approximations using KL geometry and constrained optimization.

Adaptive Client Clustering and Selection for Multi-Task Federated Learning

Research Project

- Developed CoDa-FL, combining Earth Mover's Distance (EMD)-based client clustering with Proximal Policy Optimization (PPO)-based scheduling of precedence-constrained federated learning tasks.
- Derived convergence bounds relating cluster-level EMD to communication-round requirements, providing a basis for heterogeneity-aware clustering and task-cluster assignment.
- Extended CoDa-FL to A-CoDa through balanced clustering and FedMIX, a loss- and latency-aware algorithm combining within-cluster selection with cross-cluster probing and temporary client recruitment.
- Analyzed convergence with adaptive client participation under smoothness and Polyak–Łojasiewicz assumptions, characterizing sampling-induced bias and loss-driven stochastic variance in finite-time and steady-state error bounds.
- Evaluated four dependency-constrained classification tasks in 100-client simulations, including ablations of within- and cross-cluster selection; observed a 34.29% reduction in total completion time relative to Inter-FL, the cross-cluster-only variant.

A Comparative Trade-off Analysis on Accuracy and Efficiency in Demand Forecasting

Research Project

- Conducted a comparative study of centralized, distributed and federated learning for category-level demand forecasting using daily sales data from ten Walmart stores across three U.S. states.
- Implemented multilayer perceptron models with FedAvg and stochastic federated learning (SFL), alongside ARIMA and SARIMA benchmarks under centralized and distributed learning.
- Formulated a bi-objective optimization framework balancing forecasting error and data-transmission volume, and evaluated participating-store configurations under varying objective weights.
- Characterized the trade-off between reduced per-round communication and additional convergence rounds under stochastic client participation through repeated forecasting experiments.

Hierarchical Memory Mamba for Long-Context Modeling

Research Project

- Identified the fixed-capacity recurrent-state bottleneck in Mamba-style state space models and developed **Hierarchical Memory Mamba (HMM)** for long-context modeling.
- Designed a hierarchical memory mechanism combining semantic segmentation, lightweight working memory, compressed long-term memory, and Top- K similarity retrieval.
- Conducted large-scale experiments with Mamba2 backbones on Passkey Retrieval, PG-19 and LongBench-E, including hyperparameter search, ablation studies, GPU memory, throughput and latency analysis.
- Achieved substantial improvements in long-context retrieval and reasoning while introducing only limited additional parameters.

Energy Efficient Power Control for Over-the-Air Federated Learning

Bachelor's Dissertation

- Formulated transmission-energy minimization for over-the-air federated learning in low-Earth-orbit satellite networks, subject to transmit-power, communication-rate and learning-accuracy constraints.
- Derived convergence bounds under smoothness and strong convexity, linking transmit power and aggregation error to communication-round requirements and an upper bound on total transmission energy.
- Developed OPCO, an adaptive power-control heuristic based on a relaxed per-round objective and local search near the minimum feasible power boundary.
- Evaluated OPCO using ResNet9 on non-IID CIFAR-10 partitions in a simulated network of 15 satellites; observed a 24.8% reduction in total transmission energy relative to OTA-MTP at a 70% test-accuracy target.

PUBLICATIONS AND PREPRINTS

Jieping Luo, Qiyue Li, et al. "Cluster-Based Client Selection for Dependent Multi-Task Federated Learning in Edge Computing." *arXiv preprint arXiv:2510.13132*, 2025.

Jieping Luo*, Qinwen Wang*, et al. "Mamba with Hierarchical Memory: Solving Representation Bottleneck in Long Sequence Modeling." *arXiv preprint arXiv:2608.02347*, 2026.

Hang Qi, **Jieping Luo**, Qiyue Li, and Jingjin Wu. "A Comparative Trade-off Analysis on Accuracy and Efficiency for Federated Learning in Demand Forecasting." *Applied Soft Computing*, vol. 182, 113561, 2025.

Hang Qi, **Jieping Luo**, Zimu Xu, et al. "Energy Efficient Power Control for Over-the-Air Federated Learning in Satellite Communications." *IEEE Communications Letters*, vol. 29, no. 7, pp. 1530–1534, 2025.

Qiyue Li, Yingxin Liu, Hang Qi, **Jieping Luo**, et al. "Adaptive Client Selection via Q-Learning-based Whittle Index in Wireless Federated Learning." *2025 IEEE GLOBECOM Workshops (GC Wkshps)*, pp. 2035–2040, 2025.

Hang Qi, Qiyue Li, Yuxuan Chen, **Jieping Luo**, et al. "FedPRADA: Federated Learning with Personalization, Regularization and Adaptively Decomposed Aggregation for Heterogeneous Classification with Two Shipment Data Case Studies." *Applied Soft Computing*, vol. 201, 115696, 2026.

Preprints Under Review

Jieping Luo, Qiyue Li, Yuxuan Chen, Hang Qi, et al. "Adaptive Client Clustering and Coordination for Dependent Federated Learning at the IoT Edge." *Under review at IEEE Internet of Things Journal*, 2026.

Hang Qi, **Jieping Luo**, Qiyue Li, Yuxuan Chen, et al. "Latency-Aware Topology Optimization for Hierarchical Over-the-Air Federated Learning in Non-Terrestrial Networks." *Manuscript under review*, 2026.

Qiyue Li, Hang Qi, **Jieping Luo**, et al. "Client Scheduling for Deadline-Constrained Wireless Federated Learning under Client-Level Differential Privacy." *Manuscript under review*, 2026.

Qiyue Li, Xinyi Zhou, Hang Qi, **Jieping Luo**, et al. "REFRESH: Reliable and Fresh Mobile Federated Learning for Communication-Budgeted Wearable Systems." *Manuscript under review*, 2026.

RELEVANT RESEARCH SKILLS

- **Theoretical analysis:** Derivation of convergence, finite-time and steady-state error bounds, together with analyses of asymptotic growth rates, expected stopping times and finite-sample power.
- **Mathematical modelling:** Formulation of resource-constrained learning problems involving data heterogeneity, latency, communication, energy and task dependencies.
- **Algorithm design:** Development of adaptive clustering, client scheduling, power-control and hierarchical-memory methods for distributed and long-context learning.

- **Empirical evaluation:** Design of controlled simulation studies with baseline comparisons, ablation and sensitivity analyses, hyperparameter search and accuracy–efficiency trade-off evaluation.

AWARDS AND SCHOLARSHIPS

- **Outstanding Graduate Scholarship**, 2025 — awarded to only 4 students university-wide (approx. RMB 100,000).
- **National Scholarship**, 2023–2024.
- **President’s Scholarship**, 2022–2024 — received for two consecutive years; only 10 students university-wide each year (approx. RMB 90,000 in total).
- **Dean’s Scholarship**, 2021–2022 — only 20 recipients university-wide (approx. RMB 45,000).

TECHNICAL SKILLS

Programming	Python, SQL, Bash, $\LaTeX$
Machine Learning	PyTorch, Scikit-learn, NumPy, Pandas
Deep Learning	Transformers, Mamba / State Space Models, Knowledge Distillation, Federated Learning
Systems	Linux, Multi-GPU Training, Git, GPU Profiling
Statistics	Statistical Inference, Sequential Testing, Optimization, Time-Series Modeling